

Time: 180 Minutes

Max. Marks: 50

- Instruction:**
1. In case of any doubt, write your assumptions, write it clearly and continue.
 2. There are 06 questions printed on the paper. Attempt any 5 questions.
 3. Please write all sub-parts of a question in a sequence.

- Q1. (a) Why loss function is chosen as squared loss instead of absolute or actual loss in regression problems? **02**
(b) Design a McCulloch Pitts neuron to model a 3-Bit logical AND problem. **02**
(c) Explain the weight updating equations for the following algorithms: **04**
i) Adagrad ii) RMSProp
(d) What are the characteristics of a sequential learning problem? **02**
- Q2. (a) Which of the following is/are true? If any statement is false, make it true without diverting from the context of the question. **04**
a. Simple models has high bias and high variance.
b. High bias and low variance are the desirable characteristic of a good model.
c. Variance contribute to error while Bias do not contribute to error.
d. In NAG optimizer, gradient term is first added to the old weight and then momentum term is added.
(b) Consider a convolutional neural network block whose input size is $64 \times 64 \times 8$. The block consists of the following layers: **06**
• A convolutional layer 32 filters with height and width 3 and 0 padding which has both a weight and a bias (i.e. CONV3-32)
• A 2×2 max-pooling layer with stride 2 and 0 padding (i.e. POOL-2)
• A batch normalization layer (i.e. BATCHNORM)
Compute the output activation volume dimensions and number of parameters of the layers. You can write the activation shapes in the format ($H; W; C$ where $H; W; C$ are the height, width, and channel dimensions, respectively.
i) What is the output activation volume dimensions and number of parameters for CONV3-32?
ii) What is the output activation volume dimensions and number of parameters for POOL2?
iii) What is the output activation volume dimensions and number of parameters for BATCHNORM?
- Q3. (a) Given a recurrent neural network for a sequential problem with k -dimensional inputs, n -dimensional states and d -classes. If U is the weight matrix between input to states, V is the weight matrix between states to outputs and W is the weight matrix between states then what is the dimension of U , V and W ? Mention the equations of state s_i and output y_i . **03**
(b) You are using an Adam optimizer. Show why the bias correction naturally disappears when the numbers of steps to compute the exponential moving averages gets large. **03**
(c) What are the Gates used in standard LSTM model? Explain the Gates mathematically. Why GRU is computationally cheaper than LSTM? **04**
- Q4. (a) Why an activation function is used in neural network? What are the characteristics of a good activation function? Explain the merits and drawbacks of sigmoid activation function by computing its derivative? **04**
(b) You decide to build an image classifier to predict whether a watermelon is sweet (1) or not (0). **06**
i) You've built your own labeled dataset, model and are thinking about using the mean squared error (MSE) loss to optimize model parameters. Give one reason why MSE might not be a good choice for your loss function. [1 Mark]
ii) You decide to use the binary cross-entropy (BCE) loss function. Write down the formula for this loss (for a single example) in terms of the label y and prediction y' . (1 formula) [1 Mark]
iii) What value does the loss take for a prediction of $y' = 0.9$ on a negative ($y = 0$) example. You can simply fill in the formula with the numbers, without needing to calculate it. (1 formula). [1 Mark]

iv) Compute the total cost, J , of the network averaged across the following dataset of 3 examples using the binary cross entropy loss. $Y^T = (1, 0, 0)$, and $Y^T = (0.1, 0.2, 0.7)$. (1 formula). [3 Marks]

Q5. (a) You come up with a CNN classifier. For each layer, calculate the number of parameters and the dimension of the associated feature maps (input). The notation follows the convention: **05**

- CONV- K - N denotes a convolutional layer with N filters, each them of size $K \times K$, Padding and stride parameters are always 0 and 1 respectively.
- POOL- K indicates a $K \times K$ pooling layer with stride K and padding 0.
- FC- N stands for a fully-connected layer with N neurons.

Layer	Feature map dimension	Number of parameters
INPUT	$128 \times 128 \times 3$	0
CONV-9-32		
POOL-2		
CONV-5-64		
POOL-2		
CONV-5-64		
POOL-2		
FC-10		
FC-3		

- (b) What is the number of perceptron units that are sufficient to model the XOR function with 3 variables with only a single hidden layer and an output layer? Draw the model. **02**
- (c) You want to predict the next character in an auto completion sequential problem. You are using a recurrent neural network model to solve it. However, your model don't give better accuracy. **03**
- Will shuffling your dataset improve the accuracy? Justify your answer in a single line?
 - What loss function will you choose for this problem?
 - What can you tell about the dimension of input sequence instances?

Q6. (a) The convolution operation is applied on an input image (as in Fig.1) using a filter as in Fig.2 with stride 1. What will be the output image? If max pooling is applied on the convoluted image with a 2×2 filter and stride 1 what will be its resultant feature representation? If padding is to be used on the convoluted image to make its dimension same as input image what will be the padding size? **05**

1	2	3	4	0	6
3	0	5	2	1	8
1	1	7	3	8	1
0	2	0	3	4	0
1	3	0	5	2	1
0	2	1	0	3	0

Fig. 1 Input Image

1	2	1
0	0	0
-1	-2	-1

Fig.2 Filter

- (b) Explain guided Backpropagation algorithm to vizualize the inference of a neuron in convolutional neural network. **03**
- (c) Mention at least eight forms of regularization that can be used in deep learning. **02**

*****End of Question Paper*****

NATIONAL INSTITUTE OF TECHNOLOGY ROURKELA
END - SEM EXAMINATION, 2025

SESSION: 2024 – 2025 (Spring)
M.Tech. 2nd Semester and Ph.D

Subject code: CS6592
No. of pages: 2

Subject Name: Applied Deep Learning
Full Marks: 50

Dept. Code: CS
Duration: 3 Hrs

Answer ALL questions. Unnecessary and extra writing will attract negative marks.

Q. No.	Questions	Marks	CO	BL
1. (a)	A simple RNN cell computes its hidden state using tanh activation function. Given the following parameters for the RNN cell: Input at time t: $x_t = [0.6, -0.2]$, Previous hidden state: $h_{t-1} = [0.1, 0.4]$, Input-to-hidden weight matrix: $W = \begin{bmatrix} 0.3 & 0.2 \\ 0.1 & -0.3 \end{bmatrix}$, Hidden-to-hidden (recurrent) weight matrix: $U = \begin{bmatrix} 0.4 & -0.1 \\ 0.2 & 0.5 \end{bmatrix}$, Bias vector: $b = [0.0, 0.1]$. Compute the next hidden state h_t .	5+5	CO1	BL2
(b)	Consider a simplified LSTM cell with an input dimension of 2 and a hidden dimension of 2. For the input gate, you are given: • Input $x_t = [0.5, -0.2]$ • Previous hidden state $h_{t-1} = [0.1, 0.3]$ • Weight matrix for the input gate: $W_i = \begin{bmatrix} 0.2 & 0.1 \\ 0.0 & 0.3 \end{bmatrix}$ • Recurrent weight matrix: $U_i = \begin{bmatrix} 0.1 & 0.0 \\ 0.2 & 0.2 \end{bmatrix}$ • Bias: $b_i = [0, 0]$ Compute the pre-activation value for the input gate and then apply the sigmoid function to obtain the input gate activation.			
2. (a)	An input image has dimensions 32×32 with 3 channels. It is passed through a convolutional layer with 10 filters, each size 5×5, using a stride of 1 and padding of 2. Calculate the output dimensions of the feature maps. Explain the backpropagation of CNN.	5+5	CO2	BL1
(b)	Consider the first convolutional layer of the AlexNet architecture. If the input image size is 224×224 and the layer uses a filter with a stride of 4 and no padding, determine the dimensions of the resulting feature maps. Briefly discuss the architecture of AlexNet.			
3. (a)	Consider a simplified GRU cell. The GRU computes its update gate z_t and reset gate r_t using a sigmoid activation function. Hidden state is calculated using tanh activation function. Given the following parameters: Input: $x_t = [0.3, -0.1]$ Previous hidden state: $h_{t-1} = [0.2, 0.5]$ For the update gate:	5+5	CO2	BL3

	$W_z = \begin{bmatrix} 0.4 & -0.2 \\ 0.1 & 0.3 \end{bmatrix}, U_z = \begin{bmatrix} 0.2 & 0.1 \\ -0.3 & 0.4 \end{bmatrix}, b_z = [0.0, 0.1]$ Calculate the update gate vector z_t. Show all steps. (b) Given a query vector $Q = [1, 0]$ and two key-value pairs: • Key $K_1 = [0.5, 0.5]$, Value $V_1 = [2, 1]$ • Key $K_2 = [-0.5, 0.5]$, Value $V_2 = [0, 3]$ Use the scaled dot-product attention mechanism with scaling factor $\sqrt{d_k}$ (assume $d_k = 2$) to: i) Compute the dot products between Q and each key, then scale them. ii) Apply the softmax function to obtain attention weights. iii) Compute the weighted sum of the value vectors to produce the final output. Show the complete calculations.			
4.	(a) Describe the concept of self-attention and positional encoding in Transformer models. How do these mechanisms help in capturing long-range dependencies in sequences? (b) Consider a simple autoencoder with input dimension 4 and hidden dimension 2. Encoder weight matrix $W_e = \begin{bmatrix} 1 & 0 & -1 & 2 \\ 0 & 1 & 1 & -1 \end{bmatrix}$, bias $b_e = [1, 0]^T$ and decoder weight matrix $W_d = \begin{bmatrix} 2 & -1 \\ 1 & 0 \\ 0 & 1 \\ -1 & 2 \end{bmatrix}$, bias $b_d = [1, 0, 0, 1]^T$. A single training example is $x = [1, 2, -1, 0]^T$. i) Calculate encoder and decoder output. ii) Compute the mean squared reconstruction error.	5+5	CO3	BL2
5.	(a) A U-Net, used for image segmentation, takes an input image of size 128×128. Explain with a numerical example using a max-pooling layer with a 2×2 window and stride 2, how the down sampling reduces the spatial dimensions. Explain briefly the Q-learning algorithm with the Bellman equation. (b) A simple Generative Adversarial Network (GAN) is trained on a batch size of 3 generated (“fake”) samples. For each fake sample, the discriminator outputs the probabilities of being “real” are (0.7, 0.4, 0.9). Whereas for the three “real” samples, the discriminator outputs are (0.8, 0.6, 0.3). Use binary cross-entropy loss to calculate: i) Total discriminator loss over both batches (real + fake), averaging over all 6 samples. ii) Total generator loss over the fake batch size of 3 samples, averaging over those 3 samples.	5+5	CO1	BL3

CO: Course Outcome and BL: Bloom’s Level

***** ALL THE BEST *****



NATIONAL INSTITUTE OF TECHNOLOGY ROURKELA-769008
END-SEMESTER EXAMINATION, 2024
SESSION: 2023 – 2024 (Spring)
 B. Tech. 8th / M. Tech. 2nd Semester

Subject code: CS6404 **Subject Name:** Software Testing
No. of pages: 3 **Full Marks:** 100

Dept. Code: CS
Duration: 3 Hours

Figures at the right-hand margin indicate marks.
All parts of a question should be answered at one place.
Answer any FIVE questions including Q1. Justify your answers.

Q. No.	Particulars	Marks
1		2*10=20
(a)	What will be the total number of test sessions required using pair-wise integration testing, if a call graph has total 20 nodes and 19 edges? Justify your answer.	
(b)	If a test suite T contains 100 tests of which 40 are modification-revealing for program P and modified program P', and the technique M selects 24 of these 40 tests, what is the inclusiveness of M relative to P, P' and T?	
(c)	Suppose in a design hierarchy, there are 20 numbers of modules / nodes, out of which 10 are, leave nodes. How many numbers of stubs will be required for the subordinate modules, in top down integration testing?	
(d)	What do you mean by relevant slice? Why it is so important in software testing?	
(e)	What is the implication of dynamic binding in testing of object-oriented software?	
(f)	Explain 2 coverage criteria used testing of O-O software using sequence diagrams.	
(g)	For class level testing, which test design pattern, will be most suitable?	
(h)	Mention the names of two tools that are used for testing of agile software.	
(i)	How do web-based applications differ from web-enabled applications?	
(j)	Differentiate between the two types of regression testing in Agile environment?	
2	Perform different types of performance testing for the software that implements the student admission process of NIT Rourkela. You may assume the suitable functionalities of your own.	12+08=20
(a)		
(b)	Perform bottom up integration procedure for the following system hierarchy. Show all the steps.	
	<pre> graph TD 1[1] --> 2a[2a] 1 --> 2b[2b] 2a --> 3[3] 2a --> 4[4] 2a --> 5[5] 2b --> 6[6] 2b --> 7[7] 4 --> 8[8] 4 --> 9[9] 5 --> 10[10] 6 --> 11[11] 7 --> 12[12] 7 --> 13[13] </pre>	

3. (a)	Suppose you have to develop a program for admitting students in a Data Science course in NIT Rourkela, subject to the following conditions: (i) Marks in Python ≥ 70 (ii) Marks in Java ≥ 60 (iii) Marks in C++ ≥ 60 (iv) Total in all three subjects ≥ 220 or Total in Java and Python ≥ 150 If the aggregate mark of an eligible candidate is more than 240, he/she will be eligible for the scholarship course; otherwise he/she will be eligible for a normal course. The program reads the name of the student and marks in the three subjects and generates the following outputs: (i) Not eligible (ii) Eligible for scholarship course (iii) Eligible for normal course Design at least 5 test cases for this program using equivalence class partition testing. (b) Consider the following function: <pre>int findminimum(int i, int j, int k) { int min; if (i < j) { if (i < k) { min = i;} else { min = k;}} else { if (j < k) { min = j;} else { min = k;}} return min; }</pre> (i) Draw the CFG for the above function. (ii) Calculate cyclomatic complexity. (iii) List all the linearly independent paths. (iv) Design the test cases from the linearly independent paths.	10+10=20																																
4. (a)	Consider a Banking Information System with the following functionalities. <table><tr><th>Serial Number</th><th>Functionality</th><th>Function ID in SRS</th><th>Test Cases</th></tr><tr><td>1</td><td>Login the system</td><td>F 3.4</td><td>T1</td></tr><tr><td>2</td><td>Check balance</td><td>F 3.5</td><td>T2</td></tr><tr><td>3</td><td>Deposit money</td><td>F 3.6</td><td>T3</td></tr><tr><td>4</td><td>Withdraw money</td><td>F 3.7</td><td>T4</td></tr><tr><td>5</td><td>Transfer money</td><td>F 3.8</td><td>T5</td></tr><tr><td>6</td><td>Update KYC</td><td>F 3.9</td><td>T6</td></tr><tr><td>7</td><td>Exit</td><td>F 3.10</td><td>T7</td></tr></table> Develop the test specifications and test incidence report for Withdraw money functionality. (b) Briefly explain use-case based testing with a suitable example. Mention the generic test requirements for use-case based testing.	Serial Number	Functionality	Function ID in SRS	Test Cases	1	Login the system	F 3.4	T1	2	Check balance	F 3.5	T2	3	Deposit money	F 3.6	T3	4	Withdraw money	F 3.7	T4	5	Transfer money	F 3.8	T5	6	Update KYC	F 3.9	T6	7	Exit	F 3.10	T7	12+08=20
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6	Update KYC	F 3.9	T6																															
7	Exit	F 3.10	T7																															

5. (a) (b)	Briefly explain the test procedure using Extended Use Case test pattern. Apply the above test procedure to the <i>Cash Replenishment</i> use case of an ATM system, and develop the test cases in a tabular manner. Briefly explain behavioral integration testing. Draw the state chart diagram for a <i>Credit Card</i> object. Generate at least 4 behavioral test cases for <i>Credit Card</i> using the state chart diagram.	10+10=20																																																																																																																									
6. (a) (b)	Why regression test prioritization is needed? Briefly explain regression test prioritization technique along with a suitable diagram and all the necessary steps. Consider a program with 10 faults and a test suite of 10 test cases, as shown in the table given below: <table><tr><td></td><td>T1</td><td>T2</td><td>T3</td><td>T4</td><td>T5</td><td>T6</td><td>T7</td><td>T8</td><td>T9</td><td>T10</td></tr><tr><td>F1</td><td></td><td></td><td></td><td></td><td>X</td><td></td><td></td><td>X</td><td></td><td></td></tr><tr><td>F2</td><td></td><td>X</td><td>X</td><td>X</td><td></td><td>X</td><td></td><td></td><td></td><td></td></tr><tr><td>F3</td><td>X</td><td>X</td><td>X</td><td>X</td><td></td><td></td><td>X</td><td>X</td><td></td><td></td></tr><tr><td>F4</td><td></td><td></td><td></td><td></td><td></td><td>X</td><td>X</td><td></td><td></td><td></td></tr><tr><td>F5</td><td></td><td></td><td>X</td><td></td><td></td><td>X</td><td></td><td></td><td></td><td>X</td></tr><tr><td>F6</td><td></td><td></td><td></td><td></td><td>X</td><td>X</td><td>X</td><td></td><td>X</td><td></td></tr><tr><td>F7</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>X</td><td></td></tr><tr><td>F8</td><td></td><td>X</td><td></td><td>X</td><td></td><td>X</td><td></td><td></td><td>X</td><td>X</td></tr><tr><td>F9</td><td>X</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>X</td></tr><tr><td>F10</td><td></td><td></td><td>X</td><td>X</td><td></td><td></td><td></td><td></td><td>X</td><td>X</td></tr></table> Calculate its APFD with the following orders of test cases. (i) TS1 = (T1, T2, T3, T4, T5, T6, T7, T8, T9, T10) (ii) TS2 = (T2, T1, T3, T4, T5, T6, T7, T8, T9, T10) (iii) TS3 = (T1, T2, T5, T4, T3, T6, T9, T8, T7, T10) What is the effect of changing the order of test cases? Which one is the best test suite in the context of detecting faults?		T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	F1					X			X			F2		X	X	X		X					F3	X	X	X	X			X	X			F4						X	X				F5			X			X				X	F6					X	X	X		X		F7									X		F8		X		X		X			X	X	F9	X									X	F10			X	X					X	X	10+10=20
	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10																																																																																																																	
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7. (a) (b) (c)	Briefly discuss unified process as an agile software development framework. With a suitable figure, briefly explain how testing is carried out in an agile software development framework. You are asked to develop a website for an inventory management system having the following features: (i) Buttons and boxes (ii) Field validation (iii) Static and dynamic contents (iv) Database functionalities (v) YouTube links and graphical contents Briefly explain the following types of web testing that you would perform for the above web application. a. Content testing b. Usability testing	5+7+8=20																																																																																																																									



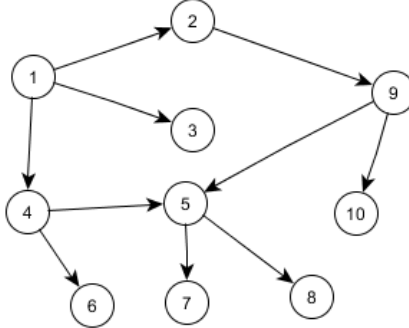
NATIONAL INSTITUTE OF TECHNOLOGY
ROURKELA – 769 008

Department	Computer Science and Engineering
Examination	End Semester – 2024-25
Program	B. Tech / M. Tech / Ph. D
Session	Spring
Course code & Course Name	CS6404 - Software Testing
Marks	50
Duration	3 Hours

This question paper contains 2 pages

Answer all questions.

Q. No.	Questions	Marks	CO*	BL**
1.	Consider the following interest calculation problem: If depositing less than Rs. 1 Lakh, rate of interest is calculated as follows: 6% for deposit upto 1 year; 7% for deposit over 1 year but less than 3 years; 8% for deposit 3 years and above. If depositing more than Rs.1Lakh, rate of interest is calculated as follows: 7% for deposit upto 1 year, 8% for deposit over 1 year but less than 3 years, 9% for deposit 3 years and above. Design test cases for the above interest calculation problem using cause-effect graph based testing technique.	5	CO-1	BL-3
2.	Consider an application <i>NameApp</i> that takes two inputs <i>name</i> and <i>age</i> where <i>name</i> is a non-empty string containing at most 25 alphabetic characters and <i>age</i> is an integer that must satisfy the constraint $0 \leq \text{age} \leq 100$. Design test cases using worst-case boundary value analysis method.	5	CO-1	BL-3
3.	Consider the following program segment: <pre>main() { int number, index; printf("Enter the number"); scanf("%d", &number); index=2; while(index<=number - 1) { if(number % index ==0) { printf("Not a prime number"); break; } index++; } if(index==number) printf("Prime number"); }</pre> Draw the CFG for the above function. Determine the cyclomatic complexity. List all the independent paths. Design test cases for all independent paths.	5	CO-2	BL-3

4.	Consider the program given below for calculating the gross salary of an employee in an organization. <pre>main() { 1. float bs, gs, da, hra = 0; 2. printf("Enter basic salary"); 3. scanf("%f", &bs); 4. if(bs < 1500) 5. { 6. hra = bs * 10/100; 7. da = bs * 90/100; 8. } 9. else 10. { 11. hra = 500; 12. da = bs * 98/100; 13. } 14. gs = bs + hra + da; 15. printf("Gross Salary = Rs. %f", gs); 16. }</pre> Find out the define-use-kill patterns for all the variables in this program.	5	CO-2	BL-3																								
5.	Suppose you have to develop a software to automate the activities of a large library. Explain the different types of performance (non-functional) testing that you will perform for the above software. You may assume your own functional and non-functional requirements.	5	CO-3	BL-3																								
6.	Perform neighborhood integration testing for the call graph shown below and calculate the effort required for this testing. 	5	CO-3	BL-3																								
7.	What do you mean by regression test prioritization? Explain the regression test prioritization technique along with a suitable diagram and the necessary steps.	5	CO-4	BL-2																								
8.	<table><tr><th>Sl. No.</th><th>Functionality</th><th>Function ID in SRS</th><th>Test Cases</th></tr><tr><td>1</td><td>Login the System</td><td>F3.4</td><td>T1</td></tr><tr><td>2</td><td>View Inventory Details</td><td>F3.5</td><td>T2</td></tr><tr><td>3</td><td>Order Book</td><td>F3.6</td><td>T3</td></tr><tr><td>4</td><td>Manage Inventory</td><td>F3.7</td><td>T4</td></tr><tr><td>5</td><td>Exit the System</td><td>F3.8</td><td>T5</td></tr></table> Develop the test specification & test log for 'View Inventory Details' functionality.	Sl. No.	Functionality	Function ID in SRS	Test Cases	1	Login the System	F3.4	T1	2	View Inventory Details	F3.5	T2	3	Order Book	F3.6	T3	4	Manage Inventory	F3.7	T4	5	Exit the System	F3.8	T5	5	CO-4	BL-4
Sl. No.	Functionality	Function ID in SRS	Test Cases																									
1	Login the System	F3.4	T1																									
2	View Inventory Details	F3.5	T2																									
3	Order Book	F3.6	T3																									
4	Manage Inventory	F3.7	T4																									
5	Exit the System	F3.8	T5																									
9.	Apply Modal Class Test Pattern to <i>DebitCard</i> class and generate the test cases. OR What is partition class testing? What are the different types of partition class testing? Generate 2 testcases for the <i>DebitCard</i> class using each type of partition class testing. You may assume your own operations / methods for the <i>DebitCard</i> class.	5	CO-5	BL-4																								
10.	Explain the agile testing life cycle with a suitable diagram. OR Suppose you are given the task of developing a web application for student admission in NIT Rourkela. Briefly explain the following types of web testing that you need to perform for the above web application. (i) Interface testing (ii) Load testing	5	CO-5	BL-2																								

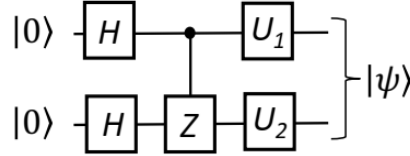
National Institute of Technology, Rourkela
End Semester Examination, Spring 2024
Quantum Computing (CS6520)

Duration: 3 Hours
Number of pages: 2

Full Marks: 50

- **Attempt any five questions.**
- **If you answer a question, all sub-questions should form part of it.**

Q 1. Consider the circuit:



- (a) Is state $|\psi\rangle$ always entangled, or does its entanglement depend on U_1 and U_2 ? Give reasons for your answer.
- (b) If U_1 is a Pauli-X gate and U_2 is a Pauli-Z gate, and the first qubit is sent to ‘Alice’, and the second to ‘Bob’, then how can Alice use the shared entangled state to send two (classical) bits of information to Bob, by transmitting only a single qubit?
- (c) If Alice knows only that U_1 is either a Pauli-X or Pauli-Z gate, but Bob knows which of these it is (and U_2 is still a Pauli-Z gate which is known to both Alice and Bob), can Alice still send two bits by transmitting only a single qubit? Explain your answer.

(2 + 4 + 4)

Q 2. Assume a flipped coin can be measured as either heads (H) or tails (T).

- (a) If the coin is in a normalized state $\frac{1}{\sqrt{10}}|H\rangle + \frac{1}{\sqrt{10}}(|T\rangle)$, what is the probability that the coin will be tails?
- (b) During a flip, the coin is in a state $\frac{1}{3}|H\rangle + \frac{2}{3}(|T\rangle)$. Is this state normalized?
- (c) A machine is built to flip coins and put them into a state $\frac{1}{2}|H\rangle + \frac{\sqrt{3}}{2}(|T\rangle)$ when flipped. If 100 coins are flipped, how many coins should land on tails?
- (d) A coin starts in the state $\frac{1}{\sqrt{10}}|H\rangle + \frac{1}{\sqrt{10}}|T\rangle$. After a measurement is made on the coin, what could be the state of the coin?

(10)

- Q 3. (a) Let $M = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & i & 0 & 0 \\ 0 & 0 & i & 1 \\ 0 & 0 & i & -1 \\ 1 & -i & 0 & 0 \end{pmatrix}$, $U_s = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$, $U_H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ and $U_{CNOT2} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$. Is the matrix M unitary? Show that the matrix M can be written as

$$M = U_{CNOT2}(I_2 \otimes U_H)(U_S \otimes U_S).$$

- (b) Let $|0\rangle, |1\rangle$ be an arbitrary orthonormal basis. Can the state

$$|\psi\rangle = \frac{1}{\sqrt{2}}|0\rangle \otimes |0\rangle + \frac{1}{\sqrt{8}}|0\rangle \otimes |1\rangle + \frac{1}{\sqrt{8}}|1\rangle \otimes |0\rangle + \frac{1}{\sqrt{4}}|1\rangle \otimes |1\rangle$$

be written as a product state?

(2 + 5 + 3)

Q 4. Let there be a database containing 32 elements, indexed by the binary numbers 00000 to 11111. A single element 00110 is marked.

- (a) Give an oracle circuit that identifies the marked element.
- (b) If Grover’s search algorithm is applied to find the marked element, what should the initial state be set to, and what is the state after a single Grover iterate has been applied?

- (c) To find the marked element with maximum probability requires N iterates in total. What is the value of N , and what is the probability of correctly finding the marked element?

(2 + 4 + 4)

- Q 5. In the superdense coding protocol, Alice and Bob each have one qubit from the entangled pair $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$, which enables Alice to send two bits of classical information to Bob, using a single qubit. Explain the superdense coding protocol in detail, and show that it does indeed enable the transmission of two bits using a single qubit.

(10)

- Q 6. (a) Write down the three-qubit Quantum Fourier Transform Matrix. Prove that such Quantum Fourier Transform Matrix is Unitary.
(b) Draw the three-qubit Quantum Fourier Transform circuit and write down the transformed states.

(5 + 5)

*******All the best*******

National Institute of Technology Rourkela
End Semester Examination, Spring 2025
Quantum Computing (CS6520/CS3012)

Duration: 3 Hours
Number of pages: 2

Full Marks: 50

- **Attempt ANY FIVE questions**

Q 1. (a) Simplify the following quantum state expression:

$$(|0\rangle \otimes I) (\alpha_{00}|0000\rangle + \alpha_{01}|0101\rangle + \alpha_{10}|1010\rangle + \alpha_{11}|1111\rangle)$$

[CO1, Apply]

(b) What is the inner product between the quantum states $|0101\rangle$ and $|0111\rangle$?

[CO1, Understand]

(c) Prove the quantum no-cloning theorem: Show that there does not exist a two-qubit unitary operator U such that

$$U(|\phi\rangle \otimes |0\rangle) = |\phi\rangle \otimes |\phi\rangle$$

for every arbitrary single-qubit state $|\phi\rangle$.

[CO2, Analyze]

(d) Show that a bit flip operation, preceded and followed by Hadamard transforms, is equivalent to a phase flip operation, i.e.,

$$H X H = Z.$$

[CO2, Analyze]

(e) Prove that an EPR pair of qubits, given by the state

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle),$$

is an entangled state, i.e., it cannot be written as the tensor product of two single-qubit states. [CO2, Analyze]

(2 + 2 + 2 + 2 + 2)

Q 2. Let $\theta \in [0, 2\pi)$, and define

$$U_\theta = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}, \quad |\phi\rangle = U_\theta|0\rangle, \quad |\phi^\perp\rangle = U_\theta|1\rangle.$$

(a) Show that $ZX|\phi^\perp\rangle = |\phi\rangle$.

[CO1, Apply]

(b) Show that an EPR pair, $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$, can also be written as

$$\frac{1}{\sqrt{2}} (|\phi\rangle|\phi\rangle + |\phi^\perp\rangle|\phi^\perp\rangle).$$

[CO2, Analyze]

(c) Suppose Alice and Bob start with an EPR pair. Alice applies U_θ^{-1} to her qubit and then measures it in the computational basis. What pure state does Bob have if her outcome is 0, and what pure state does he have if her outcome is 1?

[CO2, Analyze]

(d) Suppose Alice knows the number θ but Bob does not. Give a protocol that uses one EPR pair and 1 classical bit of communication where Bob ends up with the qubit $|\phi\rangle$ (in contrast to general teleportation of an unknown qubit, which uses 1 EPR pair and 2 bits of communication).

[CO3, Create]

(2 + 2 + 3 + 3)

Q 3. (a) Construct a CNOT gate using two Hadamard gates and one controlled-Z gate.

[CO3, Apply]

(b) Suppose our N -bit input x satisfies the following promise: either

(1) the first $\frac{N}{2}$ bits of x are all 0 and the second $\frac{N}{2}$ bits are all 1; or

(2) the number of 1s in the first half of x plus the number of 0s in the second half equals $\frac{N}{2}$.

Modify the Deutsch-Jozsa algorithm to efficiently distinguish these two cases.

[CO4, Create]

- (c) Show that for every 2×2 real unitary U , there exist signs $s_1, s_2, s_3 \in \{1, -1\}$ and angle $\theta \in [0, 2\pi)$ such that

$$U = s_1 \begin{pmatrix} 1 & 0 \\ 0 & s_2 \end{pmatrix} \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & s_3 \end{pmatrix}.$$

[CO1, Understanding]

(2 + 5 + 3)

- Q 4. **Mermin's game** is the following. Consider three space-like separated players: Alice, Bob, and Charlie. Alice receives input bit x , Bob receives input bit y , and Charlie receives input bit z . The input satisfies the promise that $x \oplus y \oplus z = 0$. The goal of the players is to output bits a, b, c , respectively, such that $a \oplus b \oplus c = \text{OR}(x, y, z)$. In other words, the outputs should sum to 0 (mod 2) if $x = y = z = 0$, and should sum to 1 (mod 2) if $x + y + z = 2$.

- (a) Show that every classical deterministic strategy will fail on at least one of the 4 allowed inputs. [CO3, Analyze]
 (b) Show that every classical randomized strategy has success probability at most 3/4 under the uniform distribution on the four allowed inputs xyz . [CO3, Analyze]
 (c) Suppose the players share the following entangled 3-qubit state:

$$\frac{1}{2}(|000\rangle - |011\rangle - |101\rangle - |110\rangle).$$

Suppose each player does the following: if his/her input bit is 1, apply H to his/her qubit, otherwise do nothing. Describe the resulting 3-qubit superposition. [CO2, Understand, Apply]

- (d) Using (c), give a quantum strategy that wins the above game with probability 1 on every input that satisfies the promise. [CO4, Create, Evaluate]

(2 + 2 + 3 + 3)

- Q 5. (a) Consider a 3-qubit state $|\psi\rangle = \frac{1}{\sqrt{8}} \sum_{k=0}^7 e^{2\pi i \cdot k \cdot 5/8} |k\rangle$.

- i. Express this state in terms of the Quantum Fourier Transform (QFT) of a computational basis state.
 ii. Apply the inverse QFT (IQFT) to $|\psi\rangle$ and write down the resulting state.
 iii. Draw the quantum circuit diagram for the 3-qubit inverse QFT using Hadamard and controlled phase shift gates.

- (b) Suppose you run Simon's algorithm for the following function $f: \{0, 1\}^3 \rightarrow \{0, 1\}^3$:
 $f(000) = f(111) = 000, f(001) = f(110) = 001, f(010) = f(101) = 010, f(011) = f(100) = 011$.
 Note that f is 2-to-1 and $f(x) = f(x \oplus 111)$ for all $x \in \{0, 1\}^3$, so $s = 111$. [CO4, Analyze]

- i. Give the starting state of Simon's algorithm.
 ii. Give that state after the first Hadamard transforms on the first 3 qubits.
 iii. Give the state after applying the oracle.
 iv. Using $H^{\otimes n}|x\rangle = \frac{1}{\sqrt{2^n}} \sum_{z \in \{0,1\}^n} (-1)^{x \cdot z} |z\rangle$, give the state after the final Hadamards.

[CO4, Analyze]

(6 + 4)

- Q 6. (a) Rewrite

$$\frac{1}{\sqrt{2}} (|0\rangle_A \otimes |0\rangle_B + |1\rangle_A \otimes |1\rangle_B) \equiv |\psi\rangle$$

in the following basis:

$$\{|+\rangle_A \otimes |+\rangle_B, |+\rangle_A \otimes |-\rangle_B, |-\rangle_A \otimes |+\rangle_B, |-\rangle_A \otimes |-\rangle_B\}.$$

What are $\text{Pr}(++)$, $\text{Pr}(+-)$, $\text{Pr}(-+)$, and $\text{Pr}(--)$? (Pr = Probability)

[CO2, Analyze]

- (b) Construct a binary adder using Toffoli gates. This adder accepts two bits X and Y , and at the output, there is a carry bit as well as $((X + Y) \bmod 2)$. [CO2, Apply]

(3 + 4 + 3)

*****All the best*****

NATIONAL INSTITUTE OF TECHNOLOGY ROURKELA

END SEMESTER EXAMINATION, 2024

MTech 2nd Semester and BTech 6th Semester

SUBJECT: Network Security

DEPT. CODE: CS 6304

FULL MARKS: 50

Number of pages: 2

Answer **Any FIVE**. Figures at the right hand margin indicate marks

All parts of a question should be answered at one place

Q.No.	Question	Marks
1	(a) In station-to-station protocol, show that if the identity of the receiver is removed from the signature, the protocol becomes vulnerable to the man-in-the-middle attack. (b) Draw block diagrams for computation of HASH1, HASH2 and HASH3 functions of IKE. The key in each case is SKEYID_d. (c) Explain why private-public keys cannot be used in creating a MAC. Distinguish between an MDC and a MAC.	[4+3+3]
2	(a) In which circumstances, a certificate must be revoked before its expiry? Discuss the certificate format with suitable diagram. (b) What do you mean by a passphrase? Why a passphrase is used in PGP? Discuss how information are retrieved at sender site in PGP with suitable diagram. (c) How the Kerberos performs the global distribution of Authentication Servers and Ticket Granting Servers?	[4+4+2]
3	(a)What content types in S/MIME provide the following security services? (i) confidentiality (ii) integrity (iii) authentication (iv) non-repudiation (v) combination of (i) and (ii) (vi) combination of (i) and (iii) (vii) combination of (i), (ii) and (iii) (viii) combination of (i), (ii), (iii) and (iv) (b) What is the length of the key material if the cipher suite is one of the following: a. SSL_RSA_WITH_NULL_MD5 b. SSL_RSA_WITH_NULL_SHA c. TLS_RSA_WITH_DES_CBC_SHA d. TLS_RSA_WITH_3DES_EDE_CBC_SHA e. TLS_DHE_RSA_WITH_DES_CBC_SHA f. TLS_DH_RSA_WITH_3DES_EDE_CBC_SHA (c) Why TLS uses PRF for all the hash calculations except for the CertificateVerify message? Justify your answer.	[4+4+2]
4.	(a) Compare the calculations of the hash for the Finished message in SSL and TLS with suitable diagrams. Which one is more secure and which one is more efficient? Justify your answer.	[4+3+3]

	(b) Show how the ISAKMP packets are exchanged between an initiator and responder using main mode with original public key method. Explain with a suitable diagram. (c) When a session is resumed with a new connection, SSL does not require the full handshaking process. Show the messages that need to be exchanged in a partial handshaking.	
5	(a) Differentiate among signed-data content type and enveloped data content type in S/MIME with suitable diagrams. (b) Which parameters are used to uniquely identify a Security Association? Give a brief introduction of each one. (c) Can a packet in PGP having tag value 6 contain another packet? Draw the format of the packet (s).	[4+4+2]
6.	(a) State the parameters used to describe the session state and connection state of SSL (b) Compare among transport mode ESP and tunnel mode ESP. (c) Which limitations of SMTP have been overcome by MIME? (d) State the benefits of IPSec.	[3+3+2+2]

NATIONAL INSTITUTE OF TECHNOLOGY ROURKELA

END SEMESTER EXAMINATION, 2025

MTech 2nd Semester and BTech 7th Semester

SUBJECT: Network Security

DEPT. CODE: CS 6304

FULL MARKS: 50

Duration of Examination: 3Hours

Number of pages: 2

Answer **Any Five questions**. Figures at the right hand margin indicate marks

All parts of a question should be answered at one place

Q. No	Question	Marks	CO	BL
1	(a) What do you mean by Cryptographic Message Syntax (CMS) in Secure MIME? Compare among Signed-Data Content Type and Enveloped-Data Content Type with suitable diagrams. (b) In which circumstances, a certificate must be revoked before its expiry? What the role of CRL in this context?	[7+3]	CO1	L1 L2
2	(a) Discuss the format of private and public key ring tables in PGP. (b) What are the various content-transfer-encoding mechanisms used in MIME protocol? Describe each one.	[7+3]	CO1	L2
3	(a) Can we use a timestamp with an asymmetric-key cipher to achieve authentication? Justify your answer. Also, discuss the operation of Fiat-Shamir protocol. (b) Discuss the importance of Key Legitimacy feature of PGP with a suitable example.	[7+3]	CO2	L3
4.	(a) In PGP, can an e-mail message use two different public-key algorithms for encryption and signing? How is this defined in a message sent from Alice to Bob? What is the length of the key material if the cipher suite is one of the following: (i). SSL_RSA_WITH_NULL_MD5 (ii) SSL_RSA_WITH_NULL_SHA (iii) TLS_RSA_WITH_DES_CBC_SHA (iv) TLS_RSA_WITH_3DES_EDE_CBC_SHA (v) TLS_DHE_RSA_WITH_DES_CBC_SHA (vi). TLS_DH_RSA_WITH_3DES_EDE_CBC_SHA (b) Define the Random Oracle Model and describe its application in analyzing attacks on hash functions.	[7+3]	CO3	L2
5	(a) What do you mean by Security Policy? What is the purpose of using Security Policy Database (SPD)? Discuss each entry of the SPD. Describe the inbound SPD processing with help of a suitable figure. (b) What do you mean by Perfect Forward Security (PFS) in context of IKE protocol? Compare the key materials created with and without using PFS.	[7+3]	CO4	L2 L3 L4

6	(a) Discuss the operation of Quick Mode of IKE protocol. Show the exchange of messages with a suitable diagram. How the three keyed-HMACs are being computed? (b) In SHA-512, show the value of the length field in hexadecimal for the following message lengths: (i) 1000 bits (ii) 10,000 bits (iii) 1000,000 bits	[7+3]	CO3 CO4	L2 L3 L4
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Department	Computer Science and Engineering
Examination	End Semester
Program	B.Tech/M.Tech//PhD
Session	Spring (2024-2025)
Course code & Course Name	Advanced Computer Networks (CS6302)
Marks	50
Duration	3 hours

This question paper contains 03 pages

Instructions:

1. Attempt all the questions, and choose any one to answer from the OR part.
2. Write answers to all parts of a question in one place.

Q. No.	Questions	Marks	CO	BL
1. (a)	Consider the following string of ASCII characters that were captured by Wireshark when the browser sent an HTTP GET message (i.e., this is the actual content of an HTTP GET message). The characters are carriage return and line-feed characters (that is, the italicized character string in the text below represents the single carriage-return character that was contained at that point in the HTTP header). Answer the following questions, indicating where in the HTTP GET message below you find the answer. <i>GET /cs6302/index.html HTTP/1.1</i> <cr> <lf> <i>Host: mcnn.cs.nitrkl.edu</i> <cr> <lf> <i>User-Agent: Mozilla/5.0 (Windows;U; Windows NT 5.1; en-US; rv:1.7.2) Gecko/20040804 Netscape/7.2 (ax)</i> <cr> <lf> <i>Accept: text/xml, application/xml, application/xhtml+xml, text/html;q=0.9, text/plain;q=0.8, image/png, */*;q=0.5</i> <cr> <lf> <i>Accept-Language: en-s,en;q=0.5</i> <cr> <lf> <i>Accept-Encoding: zip, deflate</i> <cr> <lf> <i>Accept-Charset: ISO -8859-1,utf-8;q=0.7,*;q=0.7</i> <cr> <lf> <i>Keep-Alive: 300</i> <cr> <lf> <i>Connection:keep-alive</i> <cr> <lf> <cr> <lf> (i) What is the URL of the document requested by the browser? (ii) What version of HTTP is the browser running? (iii) Does the browser request a non-persistent or a persistent connection? (iv) What is the IP address of the host on which the browser is running? (v) What type of browser initiates this message? Why is the browser type needed in an HTTP request message?	5	CO2	BL3
(b)	Consider distributing a file of $F = 10$ Gbits to N peers. The server has an upload rate of $u_s = 1$ Gbps, and each peer has a download rate of $d_i = 200$ Mbps and an upload rate of u. For $N = 10, 100$, and $1,000$ and $u = 2$ Mbps, 10 Mbps, and 100 Mbps, prepare a table giving the minimum distribution time in seconds for each of the combinations of N and u for both client-server distribution and P2P distribution.	5	CO1	BL3

2.	(a) Suppose an 802.11b station is configured to always reserve the channel with the RTS/CTS sequence. Suppose this station suddenly wants to transmit 1,000 bytes of data, and all other stations are idle at this time. Assume a transmission rate of 10 Mbps. Calculate the time required to transmit the frame and receive the acknowledgment as a function of SIFS and DIFS, ignoring propagation delay and assuming no bit errors. (b) Suppose the IEEE 802.11 RTS and CTS frames were as long as the standard DATA and ACK frames. Would there be any advantage to using the CTS and RTS frames? Why or why not? (c) In Figure 1, two wireless networks, BSS1 and BSS2, are connected through a wired distribution system (DS), an Ethernet LAN. Assume station A in BSS1 needs to send a data frame to station C in BSS2. Show the value of addresses in 802.11 frames for three transmissions: from station A to AP1, from AP1 to AP2, and from AP2 to station C. Note that the communication between AP1 and AP2 occurs in a wired environment. AP: Access Point A, B, C, D, E: stations Figure 1 <u>OR</u> Write the frame format of the baseband layer in Bluetooth (802.15.1). Describe what each of the fields signifies. Based on the frame format, answer the following questions: (i) What is the range of the address domain in a Bluetooth network? (ii) How many stations can be active at the same time in a piconet based on the information in the frame format?	3	CO4	BL3
		4	CO4	BL4
		3	CO4	BL2

3.	(a)	What is meant by the “match plus action” operation of a router or switch? In the case of destination-based forwarding packet switch, what is matched and what is the action taken? In the case of an SDN, name three fields that can be matched, and three actions that can be taken.	3	CO3	BL1
	(b)	Suppose two packets arrive to two different input ports of a router at exactly the same time. Also suppose there are no other packets anywhere in the router. (i) Suppose the two packets are to be forwarded to two different output ports. Is it possible to forward the two packets through the switch fabric at the same time when the fabric uses a shared bus? (ii) Suppose the two packets are to be forwarded to the same output port. Is it possible to forward the two packets through the switch fabric at the same time when the fabric uses a crossbar?	3	CO3	BL3
	(c)	Describe the purpose of two types of OpenFlow messages (of your choosing) that are sent from a controlled device to the controller. Describe the purpose of two types of OpenFlow messages (of your choosing) that are sent from the controller to a controlled device.	4	CO3	BL2
4.	(a)	Consider a user who needs to transmit 1.5 gigabytes of data to a server. The user lives in a village where only dial-up access (data transmission rate is 56 kbps) is available. As an alternative, a bus collects data from users in rural areas and transfers them to the Internet through a 1 Gbps link once it gets back to the city. The bus visits the village once a day and stops in front of the user’s house just long enough to receive the data. The bus has a 100 Mbps WiFi connection. Suppose the average speed of the bus is 60 km/h and that the distance between the village and the city is 150 km. What is the fastest way the user can transfer the data to the server?	3	CO1	BL3
	(b)	Suppose Alice, with a Web-based e-mail account (such as Hotmail or Gmail), sends a message to Bob, who accesses his mail from his mail server using IMAP. Discuss how the message gets from Alice’s host to Bob’s host. Be sure to list the series of application-layer protocols that are used to move the message between the two hosts.	4	CO2	BL2
	(c)	A base station can serve up to 4 users simultaneously. The arrival rate is 4 calls per minute, and average call duration is 30 seconds. (i) Calculate the offered traffic in Erlangs. (ii) Using Erlang-B, find the call blocking probability. (iii) How many calls are successfully completed per minute?	3	CO5	BL3
5.	(a)	Explain IP-in-IP encapsulation in detail. Clearly explain what each of the header fields of the encapsulated frame signifies. OR Discuss about agent discovery in Mobile-IP. Explain in detail about the agent advertisement packet?	6	CO5	BL1
	(b)	Show the interaction of mobile IP with standard TCP. Draw the packet flow from a fixed host to a mobile host via a foreign agent. Then a handover takes place. What are the following actions of mobile IP and how does TCP react?	4	CO5	BL4

CO: Course Outcome; **BL:** Bloom’s Taxonomy Level.